

ArT

Pre-Trade Controls Engine

User Manual

FINRA Rule 4210 · Regulation T · 2026 Amendments

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1. Purpose of the ArT Platform

ArT is a real-time pre-trade compliance decision engine designed to enforce regulatory requirements across equity and fixed-income trading desks. It intercepts every trade before execution and returns an immediate compliance decision with full reasoning, enabling traders, supervisors, and the controls team to act on accurate, auditable information.

2. Platform Views

The platform is organised into six views, each serving a distinct role in the pre-trade compliance workflow:

View	Role
Trade Generator	Stub to generate trades, single or in batch
Trader Entry	For Trader to manually submit a trade
Approvals	For Supervisor to override Hard Blocks
Rules Manager	For Controls Team to configure FINRA or Reg T or in-house custom rules
MI Dashboard	For Management reporting on blocks
Audit Log	For Compliance to check actions taken on the platform

3. ArT Decision Outcomes

Every trade submitted to ArT resolves to one of four outcomes:

Outcome	Meaning
Clean Pass	No rules triggered. Trade proceeds.
Soft Block	Warning issued. Trader acknowledges and can proceed.
Hard Block	Trade rejected. Regulatory hard limit — no override path.
Hard Block with Override	Trade blocked pending supervisory approval.

4. Regulatory Coverage

Regulation T (12 CFR Part 220)

- 50% initial margin for equity purchases
- 150% collateral for short sales
- Full payment for cash accounts
- Fixed income margin by credit grade

FINRA Rule 4210 (2026 Amendments)

- Intraday Margin Level (IML) computation
- Intraday margin deficit framework
- Safe harbour thresholds
- 90-day freeze enforcement
- Portfolio margin sub-\$5M check
- Pattern Day Trader (PDT) regime eliminated

5. Navigating the Platform

All views are reached from the fixed navigation menu on the left-hand side of the screen: Trade Generator, Trader, Approvals, Rules Manager, MI Dashboard, and Audit Log. Click any item to switch views — the active view is highlighted in blue. The sections below walk through each view in turn.

5.1 Trade Generator (Stub 1)

The Trade Generator is a testing stub that randomly generates trades and fires them at the ArT rules engine, covering all four outcome types across Equities and Fixed Income.

How to get here: Left-hand navigation menu → “Trade Generator” (top item, marked “Stub 1”).

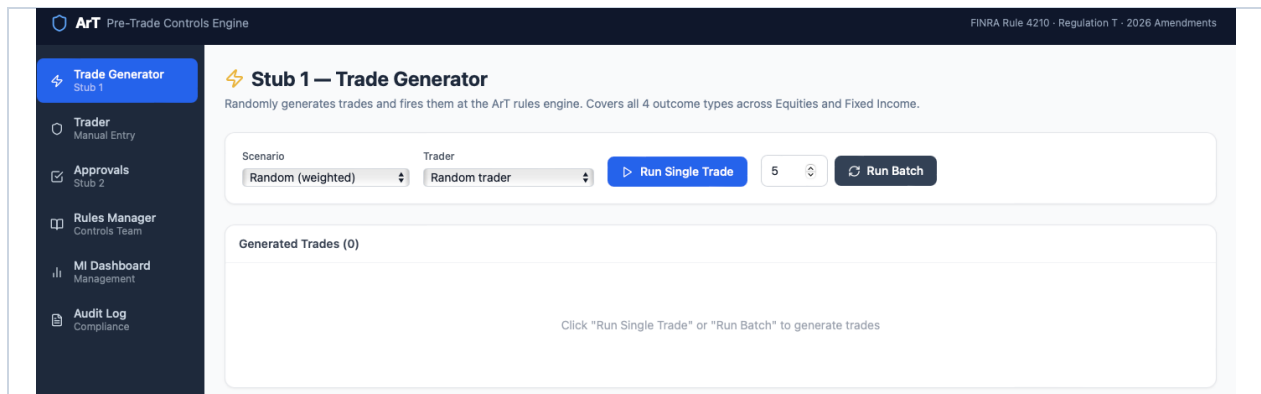


Figure 1 — Trade Generator default view, before any trades are run.

Use the Scenario dropdown to pick a specific outcome type to test (e.g. Clean Equity Buy, Hard Block Reg T, Frozen Account Short), or leave it on “Random (weighted)”.

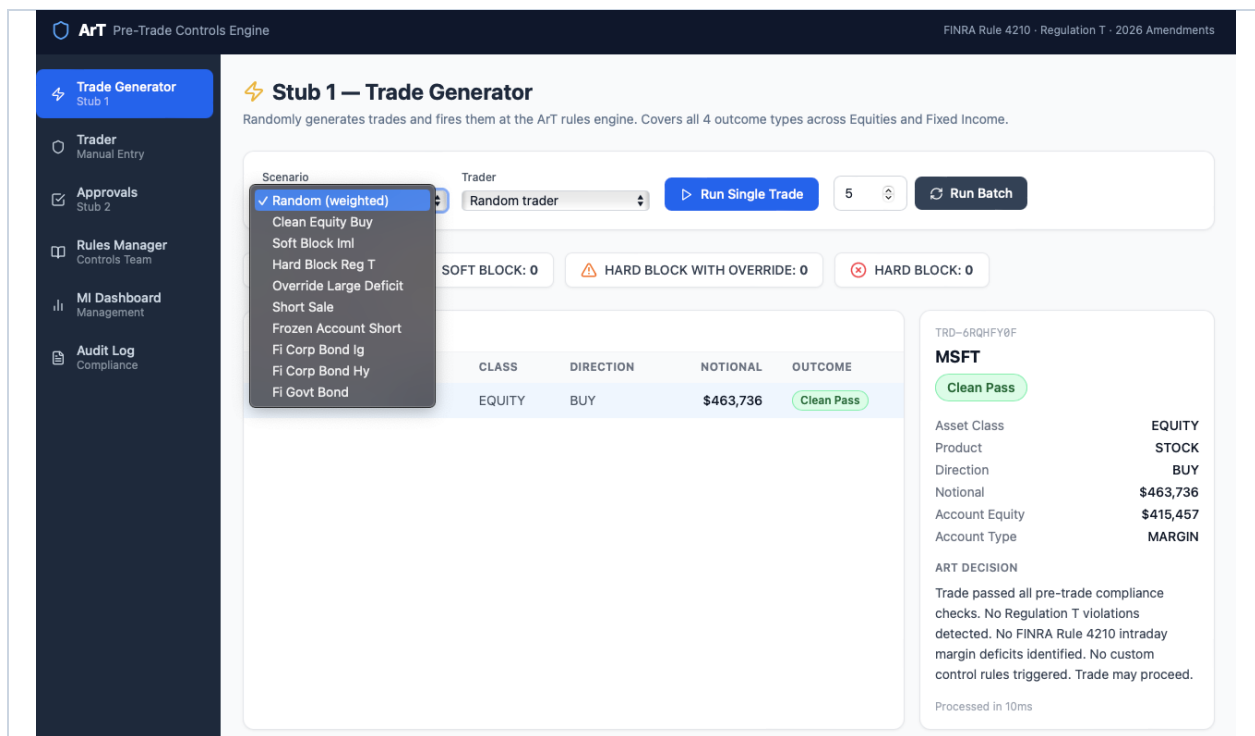


Figure 2 — Selecting a scenario from the dropdown.

Use the Trader dropdown to run the trade as a specific trader, or leave it on “Random trader”. Click “Run Single Trade” to generate one trade; the result and its full ArT Decision reasoning appear on the right.

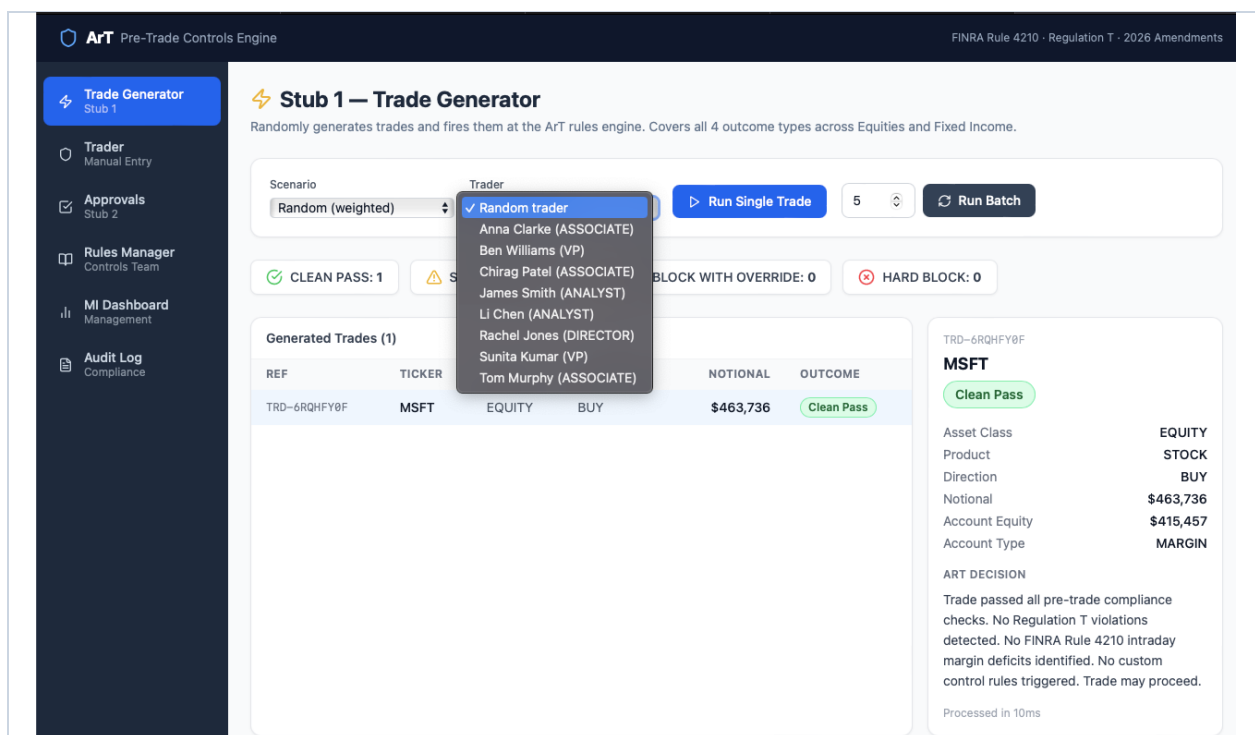


Figure 3 — Selecting a trader, and a single generated trade with its ArT decision detail on the right.

When a trade is blocked, the decision panel on the right lists every rule that is fired, with its rule code and a plain-language reason. Click any row in the trade list to inspect its decision.

The screenshot displays the ArT Pre-Trade Controls Engine interface. On the left is a dark sidebar with navigation links: Manual Entry, Approvals (Stub 2), Rules Manager (Controls Team), MI Dashboard (Management), and Audit Log (Compliance). The top control bar shows 'Scenario' set to 'Frozen Account Short' and 'Trader' set to 'Random trader'. It includes buttons for 'Run Single Trade' and 'Run Batch', along with a batch size input set to '5'. Below this, outcome counters show: CLEAN PASS: 1, SOFT BLOCK: 0, HARD BLOCK WITH OVERRIDE: 2, and HARD BLOCK: 4.

The main area features a table titled 'Generated Trades (7)' with columns: REF, TICKER, CLASS, DIRECTION, NOTIONAL, and OUTCOME. The table lists seven trades, with the first three marked as 'Hard Block' and the last two as 'Hard Block + Override'. The final trade, TRD-6RQHFY8F for MSFT, is marked 'Clean Pass'.

An arrow points from the 'Hard Block' outcome of the first trade to a detailed decision panel on the right. This panel, titled 'Hard Block', provides a breakdown of the trade details (Asset Class: EQUITY, Product: STOCK, Direction: SHORT SELL, Notional: \$989,464, Account Equity: \$791,571, Account Type: MARGIN) and the 'ART DECISION'. The decision states: 'ArT outcome: HARD BLOCK. 3 rule(s) triggered: Reg T Short Sale — 150% Collateral Requirement; 90-Day Freeze — Short Position Prohibited; Intraday Margin Deficit — Critical (Hard Block). Trade is rejected. This is a regulatory hard requirement with no override path.' Below this, a section titled 'RULES TRIGGERED (3)' lists the specific rules: 'Reg T Short Sale — 150% Collateral Requirement' (REGT-SS-001), '90-Day Freeze — Short Position Prohibited' (F4210-FRZ-001), and 'Intraday Margin Deficit — Critical (Hard Block)' (F4210-IML-004).

Figure 4 — A Hard Block outcome, showing the three rules triggered and the reasoning behind the rejection.

Set the batch size field and click “Run Batch” to generate multiple trades at once. The outcome counters at the top (Clean Pass, Soft Block, Hard Block with Override, Hard Block) update live as trades are generated.

Trader

Manual Entry

Approvals

Stub 2

Rules Manager

Controls Team

MI Dashboard

Management

Audit Log

Compliance

Randomly generates trades and fires them at the ArT rules engine. Covers all 4 outcome types across Equities and Fixed Income.

Scenario

Frozen Account Short

Trader

Random trader

Run Single Trade

5

Run Batch

CLEAN PASS: 2

SOFT BLOCK: 1

HARD BLOCK WITH OVERRIDE: 3

HARD BLOCK: 6

Generated Trades (12)

REF	TICKER	CLASS	DIRECTION	NOTIONAL	OUTCOME
TRD-3CX7CDSQ	META	EQUITY	BUY	\$418,783	Soft Block
TRD-19WM689F	GOOGL	EQUITY	BUY	\$94,813	Clean Pass
TRD-Q55HZCQU	META	EQUITY	BUY	\$4,636,213	Hard Block
TRD-AFHQ7S14	AMZN	EQUITY	BUY	\$4,929,312	Hard Block
TRD-CG8S5Z89	TSLA	EQUITY	BUY	\$261,052	Hard Block + Override
TRD-VYGRG181	BAC	EQUITY	SHORT SELL	\$989,464	Hard Block
TRD-68TJCT3	GOOGL	EQUITY	SHORT SELL	\$1,365,843	Hard Block
TRD-KKP6E9S6	TSLA	EQUITY	BUY	\$6,992,059	Hard Block
TRD-U47YVGMX	META	EQUITY	BUY	\$227,474	Hard Block + Override
TRD-156JXP2P	TSLA	EQUITY	BUY	\$2,634,124	Hard Block
TRD-8XJ7RX6N	TSLA	EQUITY	BUY	\$241,944	Hard Block + Override
TRD-6RQHFF8F	MSFT	EQUITY	BUY	\$463,736	Clean Pass

TRD-3CX7CDSQ

META

Soft Block

Asset Class

Product

Direction

Notional

Account Equity

Account Type

EQUITY

STOCK

BUY

\$418,783

\$339,286

MARGIN

ART DECISION

ArT outcome: SOFT BLOCK. 1 rule(s) triggered: Intraday Margin Deficit — Moderate. Trade carries a compliance warning. Trader should acknowledge the risk before proceeding.

RULES TRIGGERED (1)

Intraday Margin Deficit — Moderate

F4210-IML-002

Processed in 2ms

Figure 5 — Running a batch of trades; the outcome counters and trade list update for all generated trades.

5.2 Supervisory Approvals (Stub 2)

The Approvals view lists all override requests for trades that returned a Hard Block with Override outcome. Supervisors review each request — including an AI-generated risk score and rationale — and approve, reject, or escalate it.

How to get here: Left-hand navigation menu → “Approvals” (marked “Stub 2”).

Trade Generator
Stub 1

Trader
Manual Entry

Approvals
Stub 2

Rules Manager
Controls Team

MI Dashboard
Management

Audit Log
Compliance

Stub 2 — Supervisory Approvals

6 pending Refresh

Override requests for Hard Block with Override decisions. Approve, reject, or escalate with AI risk scoring.

PENDING

APPROVED

REJECTED

ESCALATED

All

REF	TRADER	REQUESTED	AI RISK	STATUS	APPROVER
OVR-WAW35K5Q	Rachel Jones DIRECTOR - Credit	16/09/2026, 09:31:20	MEDIUM 57.1/100	Pending	—
OVR-FCPH7UV1	Anna Clarke ASSOCIATE - Fixed Income	16/09/2026, 09:26:37	HIGH 69.8/100	Pending	—
OVR-VUAPFJES	Anna Clarke ASSOCIATE - Fixed Income	16/09/2026, 09:26:06	HIGH 69.9/100	Pending	—
OVR-01SLNEBU	Sunita Kumar VP - Fixed Income	25/06/2026, 01:42:36	HIGH 61.3/100	Pending	—
OVR-4VX925Q6	Li Chen ANALYST - Fixed Income	25/06/2026, 01:42:36	HIGH 76.7/100	Pending	—
OVR-RXSTTHR0	Anna Clarke ASSOCIATE - Fixed Income	25/06/2026, 01:42:34	HIGH 70/100	Pending	—

AI Risk Assessment

57.1/100 - MEDIUM

AI Rationale

Override Details

Override Ref

OVR-WAW35K5Q

Trade ID

#23

Requested

16/09/2026, 09:31:20

Take Action

Acting Supervisor *

Katherine Lee (MANAGING DIRECTOR)

Notes

Approved

Approve

Reject

Figure 6 — Pending override requests with AI risk assessment; select a request, choose the acting supervisor, add notes, then Approve or Reject.

Use the tabs above the list (Pending, Approved, Rejected, Escalated, All) to filter by status.

5.3 Rules Manager (Controls Team)

The Rules Manager lists every active pre-trade control rule — Regulation T, FINRA 4210, and custom in-house rules — with its code, type, scope, block level, and priority. New rules can be added in plain English using the AI Natural Language Rules Builder at the top of the page.

How to get here: Left-hand navigation menu → “Rules Manager” (marked “Controls Team”).

Trade Generator
Stub 1

Trader
Manual Entry

Approvals
Stub 2

Rules Manager
Controls Team

MI Dashboard
Management

Audit Log
Compliance

Rules Manager

View and manage pre-trade control rules. Use the AI builder to add new rules in plain English.

AI Natural Language Rules Builder

Powered by ArT AI

Describe a rule in plain English, e.g. "Block any equity trade where the notional exceeds \$5M and the trader is below VP level"

Parse

All Types

REG_T

FINRA_4210

CUSTOM

Active only

13 rules

CODE	NAME	TYPE	SCOPE	BLOCK	PRIORITY	STATUS
F4210-FRZ-001	90-Day Freeze — Short Position ...	FINRA 4210	ALL	Hard Block	5	Active
F4210-IML-004	Intraday Margin Deficit — Critica...	FINRA 4210	ALL	Hard Block	5	Active
REGT-CA-001	Cash Account — Full Payment R...	REG T	ALL	Hard Block	5	Active
REGT-FI-HY-001	Reg T Initial Margin — HY Corpo...	REG T	FIXED INCOME	Hard Block	10	Active
REGT-FI-IG-001	Reg T Initial Margin — IG Corpor...	REG T	FIXED INCOME	Hard Block	10	Active
REGT-SS-001	Reg T Short Sale — 150% Collat...	REG T	EQUITY	Hard Block	10	Active
REGT-LM-001	Reg T Initial Margin — Long Equi...	REG T	EQUITY	Hard Block	10	Active
F4210-IML-003	Intraday Margin Deficit — Large ...	FINRA 4210	ALL	Hard Block + Override	15	Active
F4210-IML-002	Intraday Margin Deficit — Moder...	FINRA 4210	ALL	Soft Block	20	Active

Figure 7 — The AI Natural Language Rules Builder (top) and the full rules table, filterable by type (Reg T, FINRA 4210, Custom).

5.4 MI Dashboard (Management)

The MI Dashboard gives management a real-time view of platform activity: total trades, outcome counts, a 14-day decision trend, outcome distribution, the top rules triggered, and the traders with the most block decisions. Use the date-range selector at the top right to change the reporting window.

How to get here: Left-hand navigation menu → “MI Dashboard” (marked “Management”).

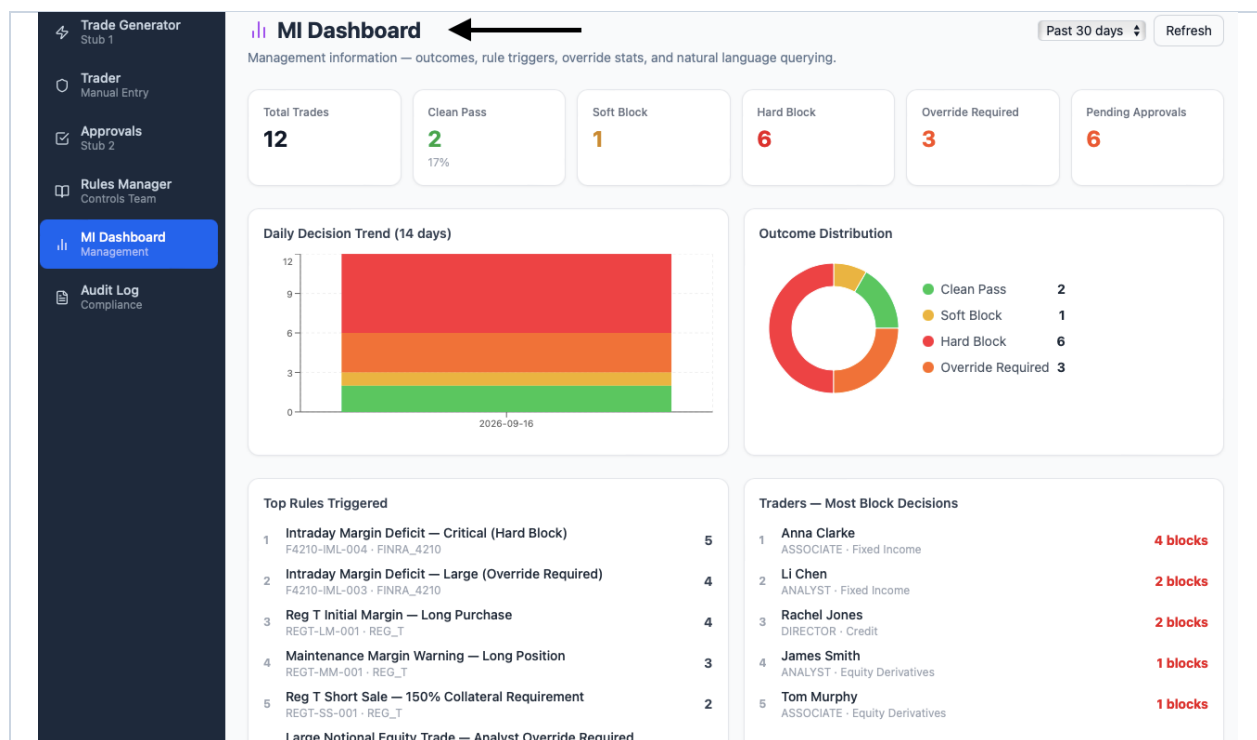


Figure 8 — MI Dashboard summary tiles, decision trend, outcome distribution, and top rules/traders tables.

5.5 Audit Log (Compliance)

The Audit Log is an immutable record of every ArT decision, override action, and system event, with timestamp, event type, related trade, acting user, and a description. Use the date-range and event-type filters at the top to narrow the log.

How to get here: Left-hand navigation menu → “Audit Log” (marked “Compliance”).

ArT Pre-Trade Controls Engine

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Trade Generator

Stub 1

Trader

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Approvals

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MI Dashboard

Management

Audit Log

Compliance

Audit Log

Immutable record of all ArT decisions, override actions, and system events.

Past 7 days

All Events

12 entries

Refresh

TIMESTAMP	EVENT	ENTITY	USER	DESCRIPTION
16/09/2026, 09:31:20	TRADE CHECKED	TRADE #23	rjones	ArT pre-trade check: HARD_BLOCK_WITH_OVERRIDE — TRD-CG8S5Z89
16/09/2026, 09:31:20	TRADE CHECKED	TRADE #22	jsmith	ArT pre-trade check: HARD_BLOCK — TRD-AFHQ7SI4
16/09/2026, 09:31:20	TRADE CHECKED	TRADE #21	tmurphy	ArT pre-trade check: HARD_BLOCK — TRD-Q55HZCQU
16/09/2026, 09:31:20	TRADE CHECKED	TRADE #20	lchen	ArT pre-trade check: CLEAN_PASS — TRD-I9WM609F
16/09/2026, 09:31:20	TRADE CHECKED	TRADE #19	aclarke	ArT pre-trade check: SOFT_BLOCK — TRD-3CX7CDSQ
16/09/2026, 09:27:12	TRADE CHECKED	TRADE #18	lchen	ArT pre-trade check: HARD_BLOCK — TRD-VYGRG181
16/09/2026, 09:27:03	TRADE CHECKED	TRADE #17	aclarke	ArT pre-trade check: HARD_BLOCK — TRD-60TWJCT3
16/09/2026, 09:26:54	TRADE CHECKED	TRADE #16	lchen	ArT pre-trade check: HARD_BLOCK — TRD-KKP6E9S6
16/09/2026, 09:26:37	TRADE CHECKED	TRADE #15	aclarke	ArT pre-trade check: HARD_BLOCK_WITH_OVERRIDE — TRD-U47YVGWX
16/09/2026, 09:26:26	TRADE CHECKED	TRADE #14	rjones	ArT pre-trade check: HARD_BLOCK — TRD-I56JXP2P
16/09/2026, 09:26:06	TRADE CHECKED	TRADE #13	aclarke	ArT pre-trade check: HARD_BLOCK_WITH_OVERRIDE — TRD-8XJ7RX6N
16/09/2026, 09:23:08	TRADE CHECKED	TRADE #12	skumar	ArT pre-trade check: CLEAN_PASS — TRD-6RQHFY0F

Figure 9 — Audit Log showing every trade check event, filterable by date range and event type.